

Progress report

1 Model

1.1 Environment

Consider a metropolitan area consisting of a finite set of locations $S = \{1, 2, \dots, n\}$. These locations are partitioned into municipalities $g \in G$:

$$S = \bigcup_{g \in G} S^g, \quad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

Each location $i \in S$ is endowed with a fixed supply of land $\bar{H}_i$, allocated between residential and commercial use by local zoning:

$$\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where $\bar{H}_i^R$ is residential floor space capacity and $\bar{H}_i^F$ is commercial floor space capacity. Both are chosen by the municipal government and assumed to be fully utilized in equilibrium.

Locations are connected by a transportation network. The generalized commuting cost from residence i to workplace j is

$$\tau_{ij} = \tau_{ij}(I_{ij}, n_{ij}),$$

where I_{ij} is transportation investment on link (i, j) and n_{ij} is the equilibrium commuter flow on that link. Investment reduces commuting costs and congestion raises them:

$$\frac{\partial \tau_{ij}}{\partial I_{ij}} \leq 0, \quad \frac{\partial \tau_{ij}}{\partial n_{ij}} \geq 0.$$

Following Fajgelbaum and Schaal, 2020, I parametrize the commuting cost as

$$\tau_{ij} = \delta_{ij}^\tau \frac{n_{ij}^\rho}{I_{ij}^\gamma},$$

where $\delta_{ij}^\tau > 0$ captures geographic features of link (i, j) and $\rho, \gamma > 0$.

There are three types of agents: households, competitive firms, and governments. Households choose residence and workplace locations. Firms produce a freely tradable good using labor and commercial floor space. Government decisions operate at two levels: municipal governments choose local zoning, and a metropolitan transportation authority chooses investment across all links. In the decentralized equilibrium, municipalities and the transportation authority choose policies simultaneously, each treating the other's decisions as given.

1.2 Households

A household chooses a residence location i and a workplace j to maximize utility. Preferences over the numaire good Q_{ij} and residential floor space h_{ij}^R are Cobb-Douglas:

$$u_{ij}(\omega) = \frac{B_i}{\tau_{ij}} \left(\frac{Q_{ij}}{\alpha} \right)^\alpha \left(\frac{h_{ij}^R}{1-\alpha} \right)^{1-\alpha} v_{ij}(\omega),$$

where $B_i > 0$ is a residential amenity at location i , $\alpha \in (0, 1)$ is the expenditure share on the composite good, and $v_{ij}(\omega)$ is an idiosyncratic preference shock drawn independently for each (i, j) pair.

The household's budget constraint is

$$Q_{ij} + q_i^R h_{ij}^R \leq w_j - \phi(N_i, L_i),$$

where q_i^R is the residential rent at location i , w_j is the wage at workplace j . Optimal demands are $Q_{ij}^* = \alpha Y_{ij}$ and $h_{ij}^{R*} = (1 - \alpha) Y_{ij} / q_i^R$. Substituting into the utility function yields the deterministic component of indirect utility:

$$\tilde{V}_{ij} = \frac{B_i (w_j - \phi(N_i, L_i))}{\tau_{ij} (q_i^R)^{1-\alpha}}.$$

The idiosyncratic shock $v_{ij}(\omega)$ is drawn from a Fréchet distribution with $\Pr(v_{ij} \leq z) = \exp(-T_i E_j z^{-\epsilon})$, where $T_i > 0$ and $E_j > 0$ are location-specific scale parameters and $\epsilon > 1$ governs the dispersion of idiosyncratic tastes. By the Fréchet aggregation property, the probability that a household chooses residence i and workplace j is

$$\pi_{ij} = \frac{T_i E_j \tilde{V}_{ij}^\epsilon}{\Phi}, \quad \Phi \equiv \sum_{i'} \Phi_{i'}, \quad \Phi_i \equiv \sum_j T_i E_j \tilde{V}_{ij}^\epsilon, \quad (1)$$

and expected utility is

$$U = \Gamma(1 - \frac{1}{\epsilon}) \Phi^{1/\epsilon}. \quad (2)$$

Let N denote total metropolitan population, fixed in a closed economy. Equilibrium residential population at each location is

$$N_i = N \sum_j \pi_{ij}.$$

Residential floor space market clearing at location i pins down the equilibrium rent:

$$q_i^R = \frac{(1 - \alpha) \sum_j N \pi_{ij} Y_{ij}}{\bar{H}_i^R}. \quad (3)$$

1.3 Firms

At each location j , competitive firms produce a freely tradable good using labor L_j and commercial floor space H_j^F :

$$Y_j = A_j L_j^\gamma (H_j^F)^{1-\gamma},$$

where $A_j > 0$ is location-specific productivity and $\gamma \in (0, 1)$. Taking wages w_j and commercial rents q_j^F as given, profit maximization yields:

$$w_j = \gamma A_j \left(\frac{H_j^F}{L_j} \right)^{1-\gamma}, \quad q_j^F = (1 - \gamma) A_j \left(\frac{L_j}{H_j^F} \right)^\gamma.$$

Commercial floor space is constrained by zoning: $H_j^F = \bar{H}_j^F$. Labor market clearing at j gives

$$L_j = N \sum_i \pi_{ij},$$

determined by household location choices. The disutility $\phi(N_i, L_i)$ uses L_i to denote employment at location i acting as a workplace, which is this same quantity evaluated at $j = i$.

1.4 Municipal Zoning

Land at each location competes for residential and commercial use. Each municipality $g \in G$ chooses a fee $\{t_i^F\}_{i \in S^g}$ levied on commercial land use at each location — an additional cost incurred whenever land is used for commercial rather than residential purposes, in the spirit of real-world commercial linkage fees that charge new commercial floor space to help offset the housing demand it generates. Because the same underlying land parcel can be put to either use, competitive landowners allocate it to whichever use pays the higher after-fee return; if both uses are active in equilibrium, this competition equalizes returns at the margin:

$$q_i^R = q_i^F - t_i^F. \quad (4)$$

Together with the land identity $\bar{H}_i^R + \bar{H}_i^F = \bar{H}_i$, the residential rent $q_i^R(\bar{H}_i^R)$ from (3) and the commercial rent $q_i^F(\bar{H}_i^F)$ from the firm's first-order condition — each strictly decreasing in its own floor-space argument — condition (4) pins down the equilibrium land split given the fee. Define

$$g(\bar{H}_i^R; t_i^F) \equiv q_i^R(\bar{H}_i^R) - q_i^F(\bar{H}_i - \bar{H}_i^R) + t_i^F.$$

Since q_i^R is strictly decreasing in $\bar{H}_i^R$ and q_i^F is strictly decreasing in $\bar{H}_i^F = \bar{H}_i - \bar{H}_i^R$ (so $-q_i^F(\bar{H}_i - \bar{H}_i^R)$ is itself strictly decreasing in $\bar{H}_i^R$), g is strictly decreasing in $\bar{H}_i^R$, so $g(\bar{H}_i^R; t_i^F) = 0$ has a unique solution $\bar{H}_i^R(t_i^F)$ for every fee. By the implicit function theorem,

$$\frac{d\bar{H}_i^R}{dt_i^F} = -\frac{1}{g'(\bar{H}_i^R)} > 0, \quad g'(\bar{H}_i^R) \equiv \frac{\partial q_i^R}{\partial \bar{H}_i^R} + \frac{\partial q_i^F}{\partial \bar{H}_i^F} < 0, \quad (5)$$

so a higher commercial-use fee raises equilibrium residential floor space: taxing commercial use pushes land toward residential. No supplier or developer is introduced — (4) is a no-arbitrage condition on a competitive land market, using only the rent functions the model already has, and total land $\bar{H}_i$ remains fixed.

The municipality sets $\{t_i^F\}$ to maximize the total welfare of its residents. In line with the political economy of local land use (Mast, 2024; Rollet and Weiwu, 2025), the municipal government acts as a representative of its residents, who face a tradeoff between

lower housing costs — which favor a higher fee, pushing land toward residential use — and lower congestion and fewer workers near their homes — which favor a lower fee, leaving land available for commercial use. Formally, municipality g solves

$$\max_{\{t_i^F\}_{i \in S^g}} W^g = \sum_{i \in S^g} N_i (\bar{U}_i - \phi(N_i, L_i)) \quad \text{s.t.} \quad (4), \quad \bar{H}_i^R + \bar{H}_i^F = \bar{H}_i,$$

where N_i is the equilibrium population at location i and $\bar{U}_i = \Gamma(1 - \frac{1}{\epsilon}) \Phi_i^{1/\epsilon}$ is the expected utility of a representative resident at location i , taking other municipalities' fees $\{t_{i'}^F\}_{i' \notin S^g}$ and transportation investment $\{I_{jk}\}$ as given. $\phi(N_i, L_i)$ is a disutility from residing at i , capturing two sources of local negative externalities: residential congestion from neighbors and the presence of workers near the residential location:

$$\frac{\partial \phi}{\partial N_i} > 0, \quad \frac{\partial \phi}{\partial L_i} > 0,$$

where N_i denotes residential population and L_i denotes employment at location i (defined below). I parametrize the disutility in Cobb-Douglas form:

$$\phi(N_i, L_i) = \eta N_i^a L_i^b, \quad \eta > 0, \quad a, b > 0, \quad (6)$$

where η is an overall congestion scale and a, b govern the relative importance of residential crowding versus worker influx. The marginal congestion costs are

$$\frac{\partial \phi}{\partial N_i} = \eta a N_i^{a-1} L_i^b, \quad \frac{\partial \phi}{\partial L_i} = \eta b N_i^a L_i^{b-1}, \quad (7)$$

both positive. Unlike an additively separable disutility, the two congestion sources are here mutual complements: $\partial^2 \phi / \partial N_i \partial L_i = \eta a b N_i^{a-1} L_i^{b-1} > 0$, so an additional resident is more burdensome the more workers already commute into i , and vice versa. Own-argument convexity, $\partial^2 \phi / \partial N_i^2 = \eta a(a-1) N_i^{a-2} L_i^b > 0$, requires $a > 1$, ensuring the first-order condition below has an interior solution.

Characterizing the optimum. Since $\bar{H}_i^R = \bar{H}_i - \bar{H}_i^F(t_i^F)$ is a strictly increasing, invertible function of t_i^F , it is equivalent — and algebraically simpler — to characterize the fee choice by differentiating the objective with respect to $\bar{H}_i^R$ directly, then translating back to the fee at the end. Substituting $\bar{H}_i^F = \bar{H}_i - \bar{H}_i^R$ and differentiating $\sum_{i \in S^g} N_i \bar{U}_i$ with respect to $\bar{H}_i^R$, the interior first-order condition at location $i \in S^g$ is

$$\underbrace{\bar{U}_i \frac{\partial N_i}{\partial \bar{H}_i^R}}_{\geq 0 \text{ (in-movers)}} + N_i \underbrace{\frac{\partial \bar{U}_i}{\partial q_i^R} \frac{\partial q_i^R}{\partial \bar{H}_i^R}}_{\geq 0 \text{ (rent benefit)}} + N_i \underbrace{\frac{\partial \bar{U}_i}{\partial \phi} \frac{\partial \phi}{\partial N_i} \frac{\partial N_i}{\partial \bar{H}_i^R}}_{\leq 0 \text{ (congestion cost)}} + N_i \underbrace{\left(-\frac{\partial \bar{U}_i}{\partial \phi} \frac{\partial \phi}{\partial L_i} \frac{\partial L_i}{\partial \bar{H}_i^R} \right)}_{\geq 0 \text{ (less worker influx)}} = 0. \quad (8)$$

The first term is positive: a higher fee t_i^F pushes land toward residential use, and the resulting looser residential zoning lowers rents, attracting additional residents from the rest of the metro area; each in-mover contributes $\bar{U}_i$ to the municipality's objective. The second term is positive: lower rents raise effective income for all N_i existing residents. The third term is negative: in-movers raise N_i , worsening congestion for all current residents through ϕ . The fourth term is positive: shifting land from commercial to residential reduces employment L_i and the associated worker congestion. The municipality sets the

fee where the benefit of attracting residents and lowering rents is offset by the congestion they impose.

Substituting the Cobb-Douglas marginal congestion costs (7) into the congestion-cost and worker-influx terms makes the interaction between the two channels explicit:

$$\bar{U}_i \frac{\partial N_i}{\partial \bar{H}_i^R} + N_i \frac{\partial \bar{U}_i}{\partial q_i^R} \frac{\partial q_i^R}{\partial \bar{H}_i^R} + N_i \frac{\partial \bar{U}_i}{\partial \phi} \eta a N_i^{a-1} L_i^b \frac{\partial N_i}{\partial \bar{H}_i^R} - N_i \frac{\partial \bar{U}_i}{\partial \phi} \eta b N_i^a L_i^{b-1} \frac{\partial L_i}{\partial \bar{H}_i^F} = 0. \quad (9)$$

Because $\partial \bar{U}_i / \partial \phi < 0$ (higher disutility lowers $\tilde{V}_{ij}$, hence Φ_i and $\bar{U}_i$, for every j), the third term is a product of a negative and two non-negative factors and so is ≤ 0 , and the fourth term (after the leading minus sign) is ≥ 0 , matching the signs already established above. The Cobb-Douglas cross-term makes clear that the congestion cost of loosening residential zoning at i is amplified when i already hosts substantial employment L_i ($N_i^{a-1} L_i^b$ rising in L_i for $b > 0$), and symmetrically that the worker-influx benefit of shifting land toward residential use is larger where residential population N_i is already high ($N_i^a L_i^{b-1}$ rising in N_i) — a complementarity the additively separable form could not generate.

Translating back to the fee. By the chain rule and (5),

$$\frac{dW^g}{dt_i^F} = \frac{dW^g}{d\bar{H}_i^R} \cdot \frac{d\bar{H}_i^R}{dt_i^F}, \quad \frac{d\bar{H}_i^R}{dt_i^F} = -\frac{1}{g'(\bar{H}_i^R)} \neq 0, \quad (10)$$

so $dW^g/dt_i^F = 0$ holds if and only if $dW^g/d\bar{H}_i^R = 0$, i.e., if and only if (8) (equivalently (9)) holds at the residential floor space implied by the arbitrage condition (4). Choosing the fee t_i^F therefore characterizes exactly the same interior solution as the underlying quantity choice; it is simply monotonically rescaled by $-1/g'(\bar{H}_i^R) > 0$, which does not change where the derivative vanishes.

$\bar{H}_i^F$ itself can equivalently be read as a *quota*: a cap on how much land may be used commercially. The fee and quota formulations describe the same one-dimensional municipal choice at i ; the model could equally be stated with a residential-side fee $t_i^R \equiv q_i^R - q_i^F$ (so that $q_i^F = q_i^R - t_i^R$), which is strictly decreasing (rather than increasing) in $\bar{H}_i^R$ but satisfies the same equivalence by the symmetric argument. I take t_i^F — a fee on commercial use — as the primary instrument going forward, since it maps most directly onto observed commercial linkage-fee policy and, together with Glaeser and Gyourko (2002)'s price/construction-cost wedge, onto how zoning restrictiveness is measured empirically.

1.5 Transportation Authority

The metropolitan transportation authority chooses investment $\{I_{jk}\}_{j,k \in S}$ to maximize aggregate household welfare, taking the full zoning vector $\{\bar{H}_i^R, \bar{H}_i^F\}$ as given:

$$\max_{\{I_{jk}\}} N \cdot \Gamma\left(1 - \frac{1}{\epsilon}\right) \Phi^{1/\epsilon}$$

subject to the budget constraint

$$\sum_{j,k} \delta_{jk} I_{jk} \leq K, \quad I_{jk} \geq 0,$$

where δ_{jk} is the per-unit cost of investment on link (j, k) and K is the total infrastructure budget.

Let $\mu \geq 0$ be the Lagrange multiplier on the budget constraint. The interior first-order condition for link (j, k) is

$$\frac{N}{\epsilon} \Gamma\left(1 - \frac{1}{\epsilon}\right) \Phi^{1/\epsilon-1} \frac{\partial \Phi}{\partial I_{jk}} = \mu \delta_{jk}. \quad (11)$$

The left-hand side is the marginal social benefit of investment on link (j, k) : lower commuting costs raise $\tilde{V}_{ij}$ for all commuters on that link, increasing Φ and hence aggregate welfare. The authority equates the marginal benefit per dollar of investment across all active links.

2 Empirical Facts

The model’s first-order conditions generate two testable predictions. From the municipal zoning FOC equation (8): locations with better transit access face additional upward pressure on population, which amplifies the congestion term and triggers a preemptive tightening of zoning — equivalently, a lower quota $\bar{H}_i^R$ or a lower commercial-use fee t_i^F (**Proposition 1**). From the transportation authority’s FOC equation (11): because the marginal social benefit of transit investment depends on ridership, which depends on the density of residents and jobs near the corridor, restrictive zoning that suppresses station-area density should lower the equilibrium return to transit investment (**Proposition 2**). Because t_i^F is a one-to-one relabeling of $\bar{H}_i^R$ (increasing rather than decreasing in it), both predictions are equivalently stated in terms of the quota or its fee-equivalent, so the empirical tests below, which use quantity-side proxies for zoning restrictiveness, still speak directly to the fee reading of the model.

I test both predictions using Dallas–Fort Worth (DFW) tract-level data. These tests are best understood as *equilibrium consistency checks* on the model’s mechanism, not as causal estimates of the effect of zoning on transportation outcomes; a separate identification strategy, using historical transit routes, planned-route instruments, or policy shocks to zoning authority, would be needed to establish causality. Both propositions find support within DFW, though neither can claim causal identification from the cross-sectional data, as I discuss below.

2.1 Data and Zoning Proxy

The primary proxy for zoning restrictiveness is the tract-level housing supply elasticity $\hat{\gamma}_i$ of Baum-Snow and Han (2024), available for the DFW metro area ($N \approx 590$ – $1,016$ tracts depending on the outcome). Lower $\hat{\gamma}_i$ indicates that the local housing stock responds less to demand shocks, corresponding to a tighter effective constraint on $\bar{H}_i^R$ in the model, equivalently a lower commercial-use fee t_i^F . Employment inflows L_i are constructed from LODES 2023 origin-destination flows, aggregated to the workplace tract. Transit access is measured using DART station locations (148 stations in DFW), with a tract classified as transit-adjacent if its centroid lies within 0.5 miles of a station (15 tracts) or 1 mile (86 tracts).

2.2 Proposition 1: Transit Access Triggers a Zoning-Tightening Response

The municipal FOC also implies that transit access, by raising residential attractiveness $\tilde{V}_{ij}$ and the resulting equilibrium population pressure N_i , should trigger a preemptive tightening of zoning: $\hat{\gamma}_i|_{\text{near transit}} < \hat{\gamma}_i|_{\text{far}}$. I estimate

$$\hat{\gamma}_i = \alpha + \beta_1 \mathbf{1}[\text{near DART}]_i + \beta_2 \text{pctdis}_i + \beta_3 \log(1 + L_i) + \varepsilon_i$$

on the subsample of $N = 581$ Baum-Snow tracts matched to 2020 Census geometry, of which 15 have a DART-adjacent centroid.

Table 1: DART Access and Housing Supply Elasticity				
	(1)	(2)	(3)	(4)
$\mathbf{1}[\text{near DART}]$	-0.152** (0.059)	-0.205*** (0.058)	-0.178*** (0.057)	-0.177** (0.079)
pctdis		-0.412*** (0.059)	-0.384*** (0.059)	-0.384*** (0.059)
$\log(1 + L_i)$			-0.032*** (0.007)	-0.032*** (0.007)
$\mathbf{1}[\text{near}] \times \text{pctdis}$				-0.011 (0.418)
R^2	0.011	0.087	0.118	0.118
N	581	581	581	581

Note: Standard errors in parentheses. Sample restricted to the 581 Baum-Snow tracts with matched 2020 Census geometry. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

DART-adjacent tracts have housing supply elasticities roughly 0.15–0.21 units lower than non-adjacent tracts, significant at the 1–5% level across specifications (Table 1), and the result is robust to controlling separately for CBD distance and employment concentration. The interaction between transit proximity and CBD distance is not significant, likely reflecting the small number of suburban DART-adjacent tracts in the sample. This provides within-DFW support for the model’s congestion channel: transit access, which raises local population pressure in the model, predicts tighter observed zoning.

2.3 Proposition 2: Restrictive Zoning and the Station-Area Density Base

The transportation authority’s FOC implies that the marginal social benefit of transit investment depends on ridership, which in turn depends on the density of residents and jobs near the corridor. If zoning suppresses that station-area density, the model predicts a lower equilibrium return to transit investment. I test this mechanism directly at the tract level within DFW, using the interaction between the Baum-Snow and Han housing supply elasticity and DART proximity. For now I focus on population density, but future work could explore the investment decision. I estimate

$$\log(\text{density}_i) = \alpha + \beta_1 \hat{\gamma}_i + \beta_2 \mathbf{1}[\text{near DART}]_i + \beta_3 \hat{\gamma}_i \times \mathbf{1}[\text{near DART}]_i + X_i' \theta + \varepsilon_i \quad (12)$$

where $\hat{\gamma}_i$ is the Baum-Snow and Han housing supply elasticity, $\mathbf{1}[\text{near DART}]_i$ indicates a centroid within 0.5 miles of a DART station, and X_i includes CBD distance, log median income, and race shares. The model’s density channel implies $\beta_3 > 0$: near transit, additional permissiveness in zoning should raise density by more than elsewhere, because transit access magnifies the returns to residential concentration.

Table 2: Zoning Permissiveness and Station-Area Density (Dallas, $N = 590$ tracts)

Outcome:	(1)	(2)	(3)	(4)	(5)
	Pop.	Pop.	log density 0.5 mi buffer	unit, 0.5 mi buffer	1 mi buffer
$\hat{\gamma}_i$	−1.709*** (0.153)	−2.189*** (0.132)	−2.295*** (0.133)	−2.549*** (0.132)	−2.384*** (0.138)
$\mathbf{1}[\text{near DART}]$			−0.923*** (0.231)	−0.841*** (0.229)	−0.531*** (0.139)
$\hat{\gamma}_i \times \mathbf{1}[\text{near DART}]$			1.747** (0.786)	1.507* (0.781)	1.487*** (0.368)
Controls	No	Yes	Yes	Yes	Yes
R^2	0.174	0.474	0.488	0.512	0.489
N	593	590	590	590	590

Note: Standard errors in parentheses. $\hat{\gamma}_i$ is the Baum-Snow and Han housing supply elasticity (higher = more permissive). Columns (1)–(2) show the baseline elasticity-density gradient. Columns (3)–(5) add the DART-proximity indicator and its interaction with $\hat{\gamma}_i$; column (4) uses housing-unit density as the outcome; column (5) defines “near DART” as a 1-mile buffer (86 DART-adjacent tracts vs. 15 in the 0.5-mile definition). Controls include CBD distance, log median household income, and minority-group population shares. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Columns (1)–(2) show that the direct elasticity-density gradient is negative and highly significant: tracts with lower supply elasticity (tighter zoning) are denser. This reflects an equilibrium confound rather than the model’s causal channel—central, high-demand locations in DFW are simultaneously denser and more politically restricted, and a static cross-section cannot separate these forces. Columns (3)–(5) show the interaction. The interaction coefficient $\hat{\beta}_3$ is positive and significant across all three specifications: 1.747** for population density at 0.5 miles, 1.507* for housing-unit density, and 1.487*** using the 1-mile buffer (Table 2). This means that near DART, the negative elasticity-density gradient is substantially attenuated: more permissive tracts achieve a notably higher density premium in the station area than more restrictive tracts. The implied elasticity-density slope is $-2.295 + 1.747 = -0.548$ near DART versus -2.295 far from DART (population density, column 3). The 1-mile buffer result, which covers 86 station-adjacent tracts and is therefore better powered, recovers a similarly sized interaction that is significant at the 0.1% level.

Taken at face value, the positive and significant $\hat{\beta}_3$ is consistent with the model’s density channel: restrictive zoning limits the station-area density concentration that would otherwise generate the ridership base for transit investment.

3 Working on

3.1 Problem 1

To get the intuition that zoning regulations intrinsically distort the transit investment decision, I need panel data on both zoning and transit investment. The current data on transit route has the opening data, but I don't have panel or historical zoning data. One Idea I have is to use the NHGIS historical data, if I could find, the population and density data for the tract, and then use this to proxy for the zoning restrictiveness back then. I will look into this and see if I can find a way to get historical zoning data.

3.2 Problem 2

Baum-Snow and Han (2024) provide a tract-level housing supply elasticity but worrying about the calibration part, I think I need to find a way to get the zoning restrictiveness for the tract. Rollet & Weiwu (2025) provide a way to get the zoning restrictiveness for the tract from zonecomics, but it is not free, I will look into this and see if it has the DFW data. A Glaeser-Gyourko-style zoning-tax measure (market price minus estimated marginal construction cost), applied to commercial rather than residential floor space, would be a natural additional proxy directly for the model's fee-equivalent t_i^F .

References

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